

JYOTHIRMAI KALAMKURI

AI Engineer — LLM Applications, RAG & Agentic AI | Python, FastAPI & REST APIs | Prompt Engineering | Cloud Deployment
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PROFESSIONAL SUMMARY

AI Engineer with 2 years of hands-on experience building and deploying AI/LLM-based solutions in production. Develops AI-powered applications using Python and modern backend frameworks (FastAPI, Flask), builds and integrates LLM-based solutions with REST APIs and workflows, and implements retrieval-augmented generation (RAG) with vector database search. Works with prompt engineering and model integration across multiple LLMs (OpenAI GPT, Claude, Gemini, Llama), and deploys production-ready AI systems on cloud using Git, Docker, and CI/CD. Experienced in agentic AI, function calling, MCP, and observability. AWS-certified. Immediate joiner, based in Hyderabad.

CERTIFICATIONS

- **AWS Academy Graduate — Cloud Architecting** (Amazon Web Services): infrastructure on AWS & cloud best practices.
- **AWS Academy Graduate — AWS Academy Data Analytics** (Amazon Web Services, Dec 2023).
- **AWS Academy Graduate — Cloud Foundations** (Amazon Web Services, Oct 2022).

TECHNICAL SKILLS

Languages & Backend: Python (primary — 2+ years); SQL; FastAPI, Flask; REST APIs; modern backend frameworks; modular code design, testing, documentation

LLMs & Prompt Engineering: LLM integration across multiple models (OpenAI GPT, Claude, Gemini, Llama, Hugging Face); prompt engineering; LLM evaluation frameworks

RAG & Vector Search: Retrieval-Augmented Generation (ingestion, chunking, embeddings, retrieval, reranking, citation grounding); vector databases (FAISS, Weaviate, pgvector, Chroma)

Agentic AI: Multi-agent workflows and orchestration (LangChain, LangGraph, CrewAI, AutoGen-compatible); MCP server built from scratch; function calling, tool-calling, guardrails, Agentic AI

Cloud & DevOps: AWS (EC2/ECR/S3/Lambda/IAM), Azure (App Service, Container Apps, DevOps Pipelines); Docker; Kubernetes-ready; CI/CD; Git

Data & Databases: Data and embedding pipelines; PostgreSQL, MongoDB, Redis, Supabase; SQL

Observability: Monitoring & observability (Prometheus, Grafana, Sentry); logging, tracing, cost monitoring, quality scoring

Practices: Debugging, optimization, and problem solving; Agile; Responsible AI (guardrails, human-in-the-loop)

PROFESSIONAL EXPERIENCE

AI Engineer

IICL (Intelligence India Com Limited), Hyderabad, India | IT Services | Sep 2024 – Present (promoted from AI Engineer Intern in 6 months)

- **Developed AI-powered applications using Python and modern backend frameworks** — built production AI/LLM applications (banking, ITSM, ESG compliance) with FastAPI.
- **Built and integrated LLM-based solutions with APIs and workflows** — integrated multiple LLMs (OpenAI, Claude, Gemini, Llama) into products via REST APIs.
- **Implemented retrieval-augmented generation and vector database search** — RAG with chunking, embeddings, retrieval, reranking, and citation grounding (FAISS/Weaviate/pgvector).
- **Designed and maintained REST APIs for AI services** — modular, tested, documented FastAPI services.
- **Worked with prompt engineering and model integration across multiple LLMs** — versioned prompts, multi-LLM routing, and evaluation.
- **Deployed and supported production-ready AI systems on cloud** — Docker, AWS and Azure, with CI/CD and monitoring.
- **Built agentic systems** — CrewAI/LangGraph multi-agent workflows and an MCP server from scratch with function calling.
- **Collaborated on debugging, optimization, and problem solving** — using Git, Docker, and related tooling; promoted from intern to engineer in 7 months.

PROJECTS

VeroPath — Multi-Agent ESG Compliance AI Backend

Role: AI Backend Engineer | Python, FastAPI, CrewAI, RAG, Azure

Built a CrewAI four-agent FastAPI backend that automates UAE FDL#11 regulatory ESG report generation. Implemented a RAG pipeline with citation grounding over the regulatory corpus, a versioned prompt registry, quality gates, audit logging, response caching, provider-failure fallbacks, and token/cost monitoring. Enforced human-sign-off governance. Containerised (Docker) and deployed to Azure Container Apps via Azure DevOps CI/CD across 7 user stories / 55 tasks, with a 25-model LLM catalog via Supabase.

Praneeth Ixora — Trilingual Voice AI Agent

Role: Full-Stack & Voice-AI Prompt Engineer, DevOps | Python, Next.js, Sarvam/Twilio/Gemini, Azure

Built and shipped a production English/Telugu/Hindi voice AI agent integrating three third-party platforms (Sarvam, Twilio, Gemini) via REST APIs. Authored a versioned 250-line system prompt with persona classification and guardrails, treating prompt engineering as production code. Eliminated silent version-drift via live version fetching. Passed 15/15 acceptance criteria at ~0.75s average response.

Multi-Layer AI Platform & MCP Server

Role: AI Engineer | Python, RAG, Vector DBs, LangGraph/CrewAI

Built a Model Context Protocol (MCP) server from scratch for standardized tool registration, function calling, and context passing, making agents reusable across projects. Developed a configurable RAG platform with multiple vector databases (FAISS, Weaviate, pgvector, Chroma), chunking, embeddings, and hybrid retrieval with reranking. Orchestrated multi-agent workflows with LangGraph and CrewAI.

iVaak AI — Multi-Tenant Voice Agent Platform

Role: Backend / Full-Stack Engineer | Python, LLMs, RAG, AWS

Built a production multi-tenant voice-agent SaaS (LLMs, RAG, ElevenLabs, Twilio) with Supabase auth, PostgreSQL, and row-level security, deployed on AWS with Docker and CI/CD.

EDUCATION

B.Tech, Electronics & Communication Engineering — Sreenidhi Institute of Science and Technology, 2020 – 2024